

Experiment No- 5

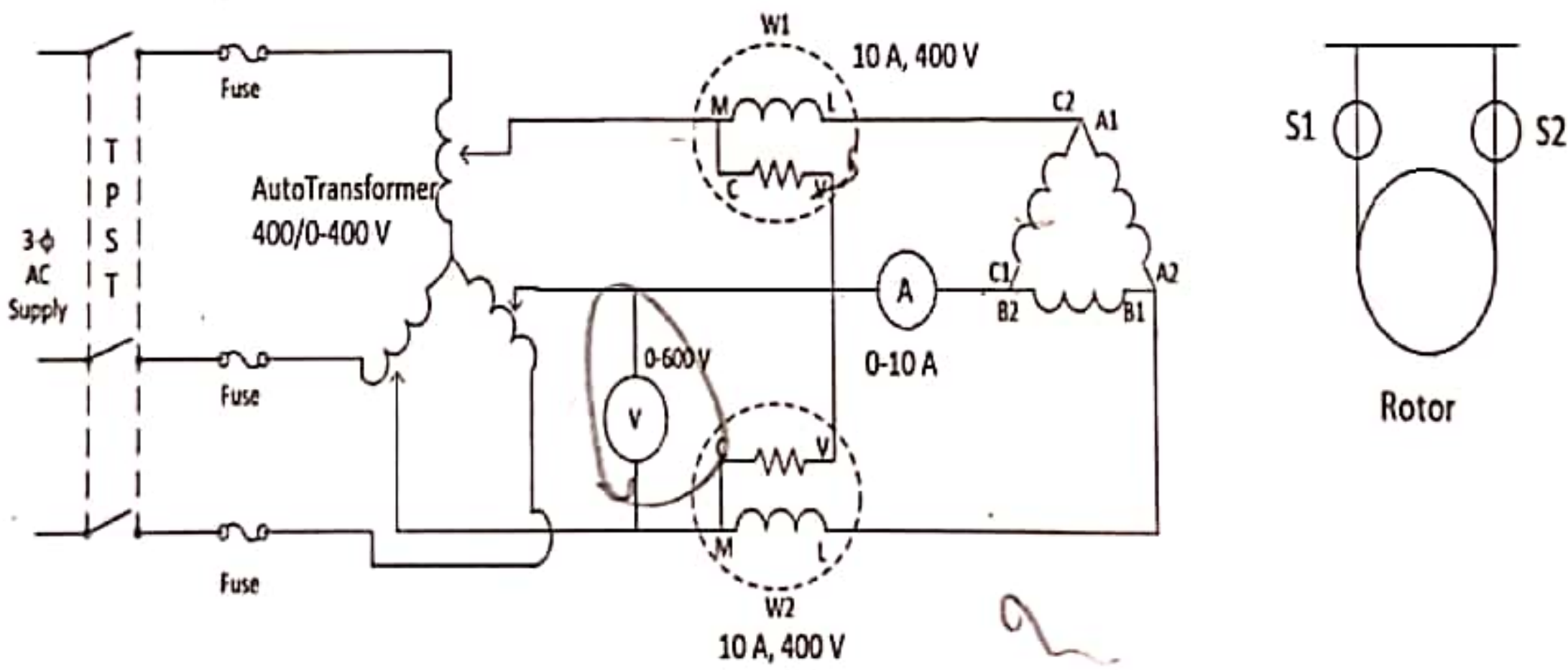
Aim:

To perform load test on 3-Ø Induction motor

Apparatus Required:

S. No.	Apparatus/Accessories	Specifications	Quantity
1	Ammeter	0-10/20A, MI	1
2	Voltmeter	0-300/600V, MI	1
3	Wattmeter	10/20A, 200/400V	2
4	3-φ Variac	15A, 400/0-400V	1
5	Tachometer	0-2000 rpm	1

Circuit Diagram:



Theory:

The load test on induction motor is performed to compute its complete performance i.e., torque, slip, efficiency, power factor etc.

In this test the motor is operated for different loads and the applied voltage, current and input power measured along with the slip of the motor. Again, the connections are made as per the circuit diagram. For all the load conditions the applied voltage should be same and equal to the rated voltage. In case the supply voltage remains constant during the experiment the auto transformer may be removed the load should be varied from a low value to an overload value the load applied to the motor may be an electrical load or a mechanical load. For electrical loading a generator (AC or DC) is coupled to the motor either directly or by belt. The output of the generator is measured. The efficiency of the generator at various loads must be known so that the output of the motor or the generator input can be calculated when the generator is coupled by a belt an allowance must be made for the loss in the belt.

In the mechanical loading some form of brake is used. A simple method consists in loading up the motor by means of a simple band brake with a spring balance on each side to measure the pull on the tight and slack sides of the belt. In another method mechanical load replaces the tight side spring balance. If S_1 is the weight applied (or the pull on the tight side spring balance) in kg and S_2 is the pull in kg on the slack side spring balance then the net force applied at the rim of the pulley is $(S_1 - S_2) \times 9.81$ in Newton and the developed torque is $T = (S_1 - S_2) \times 9.81 \times R$ N-m where R is the radius of the pulley in metre and the gravitational force g is taken as 9.81 m/s^2 . To be more accurate half of the thickness of the belt is added in the radius of the pulley. Therefore, for determining the output, speed and torque must be known. For this the speed of the motor and the values of S_1 and S_2 must be obtained. The radius of the pulley and the thickness of the belt must also be measured thus the observations for complete range of loads are obtained. Efficiencies, power factors and slips for the range of loads are calculated and these and input current measured or plotted against the output these curves are the performance curves of the motor. The disadvantage of this method of testing is that the pulley and the belt very soon get extremely hot so that the observations have to be taken quickly. Some methods of cooling the pulley should there for be used to keep the pulley and belt cool. The performance can be calculated, following the steps given below.

Slip:

The speed of rotor, N_r drops slightly as the load on the motor is increased. The synchronous speed, N_s of the rotating magnetic field is calculated, based on the number of poles p , and the supply frequency f , i.e.

$$\text{Synchronous speed } N_s = \frac{120f}{P} \text{ r.p.m}$$

$$\text{Then, slip } s = \frac{N_s - N_r}{N_s} \times 100 \text{ Percent}$$

Normally, the range of slip at full load is from 2 to 5 percent.

Torque:

Mechanical loading is the most common type of method employed in laboratories. A brake drum is coupled to the shaft of the motor and the load is applied by tightening the belt, provided on the brake drum. The net force exerted at the brake drum in kg is obtained from the readings S_1 and S_2

of the spring balances i.e.

$$\text{Net force exerted, } W = (S_1 - S_2) \text{ kg}$$

$$\begin{aligned} \text{Then, load torque, } T &= W \times d/2 \text{ kg-m} \\ &= W \times d/2 \times 9.8 \text{ Nw-m} \end{aligned}$$

where, d is the effective diameter of the brake drum in meters.

Output power, (P_o) :

The output power in watts developed by the motor is given by,

$$\text{Output power, } P_o = \frac{2\pi NT}{60} \text{ watts}$$

where, N is the speed of the motor in r. p. m.

This is the most important parameter because this will be the reference variable for each quantity to be plotted in curves.

Input power:

Input power is measured by the two wattmeters, properly connected in the circuit i.e.

$$\text{Input power} = (W_1 + W_2) \text{ watts}$$

where, W_1 and W_2 are the readings of the two wattmeters obtained after multiplying them with the respective wattmeter's multiplying factors.

Power factor:

Power factor of induction motor on no load is very low because of the high value of magnetizing current due to presence of air gap between the stator and rotor. With the increase in load, the power factor increases because the active power component of the current is increased. To reduce the magnetizing current in Induction motors the air-gap is kept as small as possible. Input power factor can also be calculated from the readings of two wattmeter for balanced load.

If p is the power factor angle, then

$$\tan^{-1} p = \sqrt{3} \frac{W_1 - W_2}{W_1 + W_2}$$

Knowing the power factor angle p , power factor $\cos(p)$ can be calculated.

Efficiency:

$$\text{Percentage efficiency of the motor, } \eta = (\text{Output power} / \text{Input power}) \times 100$$

Efficiency of a three-phase induction motor lies in the range of 75 % - 85 % under the full load condition.

Procedure:

1. Connect the circuit as per circuit diagram in figure 1.
2. Ensure that the motor is unloaded and the variac is set at zero output voltage.
3. Switch-on 3 phase ac mains and start the motor at reduced applied voltage. Increase the voltage, till its rated value.
4. Take-down the readings of all the meters and the speed.
5. Increase the load on the motor gradually by turning hand wheels. Record the readings of all the meters and the speed at every setting of the load. Observation may be continued upto the full load current rating of the motor.
6. Reduce the load on the motor and finally unload it completely.
7. Reduce the applied voltage to zero gradually then switch-off the supply to stop the motor.

Observation Table:

S. No.	V(volts)	I(Amp)	W_1 (watts)	W_2 (watts)	Speed
1					
2					
3					
4					
5					

Calculations:

Precautions:

- 1) Connections must be done as per circuit diagram.
- 2) Correct range of instruments must be connected.
- 3) Voltage applied through secondary of transformer must be kept low in the starting.
- 4) Connection of current coil of wattmeter must be reversed if reading is negative.
- 5) Current must not exceed the rated current.
- 6) Voltage must be kept constant during the load test.